

Prop. Let V be any vector space, and $T: V \rightarrow V$ be a linear map. Suppose that there exists a T -invariant subspace W such that

$$V = \text{Im } T \oplus W.$$

Then, $W \subseteq \text{Ker } T$.

Proof. Let $\alpha \in W$ be given. Then, $T(\alpha) \in \text{Im } T \cap W$, so $T(\alpha) = 0$. Thus $T(\alpha) = 0$, so $\alpha \in \text{Ker } T$.

Prop. Let V be any vector space, and $T: V \rightarrow V$ be a linear map. Then

$$\text{Im } T \cap \text{Ker } T = \{0\} \text{ if and only if } \text{Ker } T = \text{Ker } T^2.$$

Proof. Notice that: generally, $\text{Ker } T \subseteq \text{Ker } T^2$.

Suppose that $\text{Im } T \cap \text{Ker } T = \{0\}$. Let $\alpha \in \text{Ker } T^2$ be given. Then, $T^2(\alpha) = T(T(\alpha)) = 0$, so that $T(\alpha) \in \text{Im } T \cap \text{Ker } T$, so $T(\alpha) = 0$.

Thus $\alpha \in \text{Ker } T$. Conversely, suppose that $\text{Ker } T = \text{Ker } T^2$.

Given $\alpha \in \text{Im } T \cap \text{Ker } T$, $\alpha = T(\beta)$ for some $\beta \in V$ and $T(\alpha) = 0$.

Thus, $T(\alpha) = T^2(\beta) = 0$, so $\beta \in \text{Ker } T^2 = \text{Ker } T$, therefore $\alpha = T(\beta) = 0$. So, proved. $\square$

Thm. Let V be a finite-dimensional vector space, let $T: V \rightarrow V$ be linear. There is $k \in \mathbb{N}$ such that $V = \text{Im } T^k \oplus \text{Ker } T^k$.

Proof. Since $T(0) = 0$, so given $\alpha \in \text{Ker } T$, $T^2(\alpha) = T(0) = 0$, thus $\alpha \in \text{Ker } T^2$. Inductively, we obtain a chain

$$\text{Ker } T \subseteq \text{Ker } T^2 \subseteq \dots \subseteq V$$

But, since the V is finite-dimensional, for some $k \in \mathbb{N}$ such that $k \leq \dim V$,

$$\text{Ker } T^{k-1} \subsetneq \text{Ker } T^k = \text{Ker } T^{k+1} = \dots$$

Let $\alpha \in \text{Im } T^k \cap \text{Ker } T^k$ be given. Then, $\alpha = T^k(\beta)$ for some $\beta \in V$, and so that $0 = T^k(\alpha) = T^{2k}(\beta)$, so $\beta \in \text{Ker } T^{2k} = \text{Ker } T^k$, thus $\alpha = T^k(\beta) = 0$. Now, by the dimension theorem,

there exist a basis β_1 for $\text{Im } T^k$ and β_2 for $\text{Ker } T^k$ such that

$$|\beta_1| + |\beta_2| = \dim V,$$

And $\beta_1 \cup \beta_2$ is linearly independent, since $\langle \beta_1 \rangle \cap \langle \beta_2 \rangle = \text{Im } T^k \cap \text{Ker } T^k = \{0\}$, so, Consequently $V = \text{Im } T^k \oplus \text{Ker } T^k$. $\square$